

FALCON's Response to Emirates SkyCargo for the supply of Loop CBS of 9K PPH throughput.

March 16, 2023



www.falconautotech.com

Kind Attention –

Mr. Laurent

M/s Emirates SkyCargo

Offer Ref: F23-00148; Date 02.02.2023.

Subject – Techno -Commercial Offer of Loop Cross Belt Sorter of 9k PPH

Dear Sir,

We are pleased to submit our Techno-Commercial Offer in response to your RFQ

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

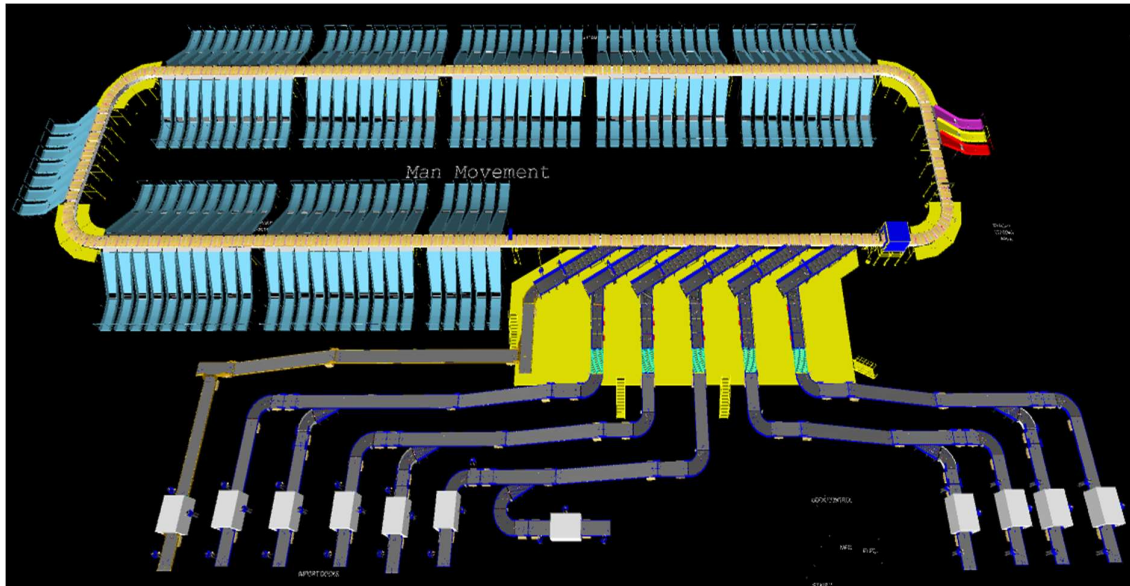
In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies and references.

To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

Contact Person : Animesh Agarwal
Title : Sales Manager
Mobile : +91-95 9980 4042
Email : Animesh.Agarwal@falconautoonline.com

Company Name : Falcon Autotech Pvt. Ltd.
Regd. Office : 57, Nimri Colony, Double Storey Flats, Delhi – 110052
Website : www.falconautotech.com
LinkedIn : [linkedin.com/in/animesh-agarwal-94b40916](https://www.linkedin.com/in/animesh-agarwal-94b40916)

Response to RFP for Cross Belt Sorter_ Emirates SkyCargo



Proposal Reference - F23-00148

Falcon Autotech Private Limited
Plot No 308 & 309
Ecotech-1 Extension, Greater Noida
Uttar Pradesh- 201308
India.

Contact – Animesh Agarwal
Sales Manager
Mob - +91 9711 4886 06
Animesh.agarwal@falconautoonline.com

Contents

1. Glossary	6
2. Company Profile	7
3. Falcon's Experience and Achievements in Sortation Space Globally	16
4. Reference projects	18
4.1 Project 1- (Ecommerce Client)	18
4.2 Project 2- (Client – Courier Express Parcel)	19
4.3 Project 3- (Client – E-Commerce)	20
4.4 Project 4- (Client – Indian Retailer)	20
5. Handled Parcel Spectrum	21
5.1 Parcel size loadable on the sorter	21
5.2 Parcels to be loaded on Sorter shall have the following characteristics:	22
5.3 Parcel not loadable on the sorter	22
5.4 Pictures of Sortable Parcels	23
6. Proposed System Description	24
6.1 Objective	24
6.2 Summary of the System	24
6.3 Main benefits of the proposed solution	25
7. Layout View	25
8. Proposed System Capacity_ Calculations	27
8.1 Sorter System Capacity	27
8.2 Infeed System	27
8.3 Feedlines	28
8.3.1 Short Buffer Conveyors	29
8.3.2 Intelligent Merge Conveyor.	30
8.4 Main Loop	31
8.5 Output Chutes	32
9. Proposed System Technical Details	33
9.1 Mechanical equipment	33
9.2 Electrical Equipment	34
9.3 Control System	34
10. Description of Components of Equipment	34
10.1 Cross Belt Sorter -	34
10.1.1 CBS Carrier	35
10.1.2 Servo Roller	36
10.1.3 Chassis	36

10.1.4	Carrier Wheels	37
10.1.5	Non-contact based linear motor drive	37
10.1.6	Power Transmission.....	37
10.1.7	Data Transmission.....	38
10.1.8	Carriers positioning System.	38
10.1.9	Technical specifications of Sorter	39
10.2	Scanning & Sensing on Main Sorter Loop	40
1.	Automatic Barcode Scanning & Dimensioning System	40
2.	Parcel Positioning System.	40
3.	Empty Carrier Detection System.....	41
10.3	Conveyor System.....	41
a.	Belt Conveyor	41
b.	Waterfalls	42
c.	Conveyor BOQ	43
10.4	Steel Works – Mezzanine, Maintenance Platforms & Staircases-	43
11.	Electrical System	44
a.	Reference Picture of Power Distribution Panel	44
b.	Main Control Panel (Reference).....	44
c.	Induct Stations Control Panel (Reference)	45
12.	Controls Architecture.....	46
13.	IT Architecture (Ref.).....	47
14.	Falcon's WCS CONTROLIT.....	47
15.	DLIC (Deep Learning Image Classification)	48
16.	Key Components Make.....	49
17.	Principle of Safety	50
18.	Additional Scope-.....	51
19.	USPs of Falcon's Cross Belt Sorter	52
20.	Automation Management System (AMS).....	53
a.	Issue Logging	54
b.	Preventive Maintenance Check List.....	54
c.	Predictive Maintenance	55
d.	Training Portal	55
e.	Operational Analysis.....	56
21.	SOPs for feeding	56
i.	Initial Checks	56
ii.	Parcel Selection.....	56

iii.	Feeding	57
22.	Program Organisation	58
23.	Price Schedule	59

1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Parcels Per Hour

3	ARB	Actuated Roller Balls
4	DBO	Damaged Barcode
5	VDS	Volume Distribution System
6	ICR	Intelligent Character Recognition
7	MEZZ	Mezzanine
8	LIM	Linear Induction Motors
9	ECDS	Empty Carrier Detection System
10	AC	Alternating Current
11	DC	Direct Current
12	PLC	Programmable Logic Controller
13	IT	Information Technology
14	BOQ	Bill Of Quantity
15	I/O	Input/ Output
16	PDP	Power Distribution Panel
17	PC	Personal Computer
18	UPS	Uninterrupted Power Supply
19	CBS	Cross Belt Sorter
20	DNF	Data Not Found
21	LGM	Logic Mismatch
22	RTVC	Real Time Video Coding

2. Company Profile

THE COMPANY

OUR VISION

Create a Global impact on the society by facilitating World's leading Organizations run their Supply Chains efficiently by leveraging technology enabled products and Solutions and, in the process, establish Falcon Autotech as one of the World's most respected and trusted brand in Intra-logistics Warehouse Automation Space.



FALCON's Proposal to Emirates SkyCargo for the supply of Loop CBS of 9k PPH Throughput

**FALCON
AUTOTECH**

COMPANY HISTORY





FALCON's Proposal to Emirates SkyCargo for the supply of Loop CBS of 9k PPH Throughput

**FALCON
AUTOTECH**

KEY STATISTICS



04

Product Lines



525+

On-Roll Employees



290+

Engineering Team



400000+ Sq Ft

Manufacturing Space



20+

Tier A Clients



15+

Countries with Live Installations



1700+

Total Installed Systems Globally



35

Representations in 4 Continents and 35 Countries



1 Million+

Sorts Per Hour of combined Sorting Capacity
installed Globally

OUR PRODUCT PORTFOLIO



SORTATION SOLUTIONS

01

- ❖ Cross Belt Sorter
- ❖ Tilt Tray Sorter
- ❖ Sliding Shoe Sorter
- ❖ Pop Up Sorter
- ❖ Swivel Wheel Sorter
- ❖ Arm Sorter
- ❖ Pusher Sorter
- ❖ Drop Down Conveyor Sorter

PICK/PUT TO LIGHT SYSTEMS

02

- ❖ Packet Sortation CEP
- ❖ Multi-Line Order Consolidation for E-Commerce
- ❖ Order Picking for Food, FMCG, Pharmaceutical
- ❖ Store Order Fulfillment for Fashion & Grocery
- ❖ Spare Part Distribution for Automobile, Electronics
- ❖ Kitting for Automotive, Consumer Electronics, Heavy Machinery Companies
- ❖ Multi Order Picking Cart



DIMENSION & WEIGHT SCANNING SYSTEMS

03

- ❖ Static Dimension & Weight Scanning Systems
 - Cubizon-R
 - Cubizon-R ECO
 - Cubizon-R THRU
 - Cubizon-R Cross
- ❖ Dynamic Conveyor Based Systems
 - Semi-Automatic Dynamic Profilers
 - Fully-Automatic Dynamic Profilers

OUR PRODUCT PORTFOLIO



CONVEYOR AUTOMATION SOLUTIONS

04

- ❖ Belt Conveyors
- ❖ Roller Conveyor
- ❖ Modular Belt Conveyors
- ❖ Inclines/ Decline Conveyors
- ❖ Conveyor Bends
- ❖ Traffic Control Merges
- ❖ Product Accumulation
- ❖ Code Scanning
- ❖ Automated Label Print And Apply
- ❖ Spirals
- ❖ In-line Weighing And Dimensioning
- ❖ In-line Carton Packaging
- ❖ In-Line Case Packaging Systems (Taping, Strapping, Shrink Wrapping)
- ❖ In Line Pouch On Demand Packaging (Cold Seal Packing)

SOFTWARE SUITE

05

❖ ControlIT - Warehouse Automation Control System

- Sorter Control Package
- DWS Control Package
- PPTL Control Package
- Dashboard Package

❖ AMS (Automation management System)

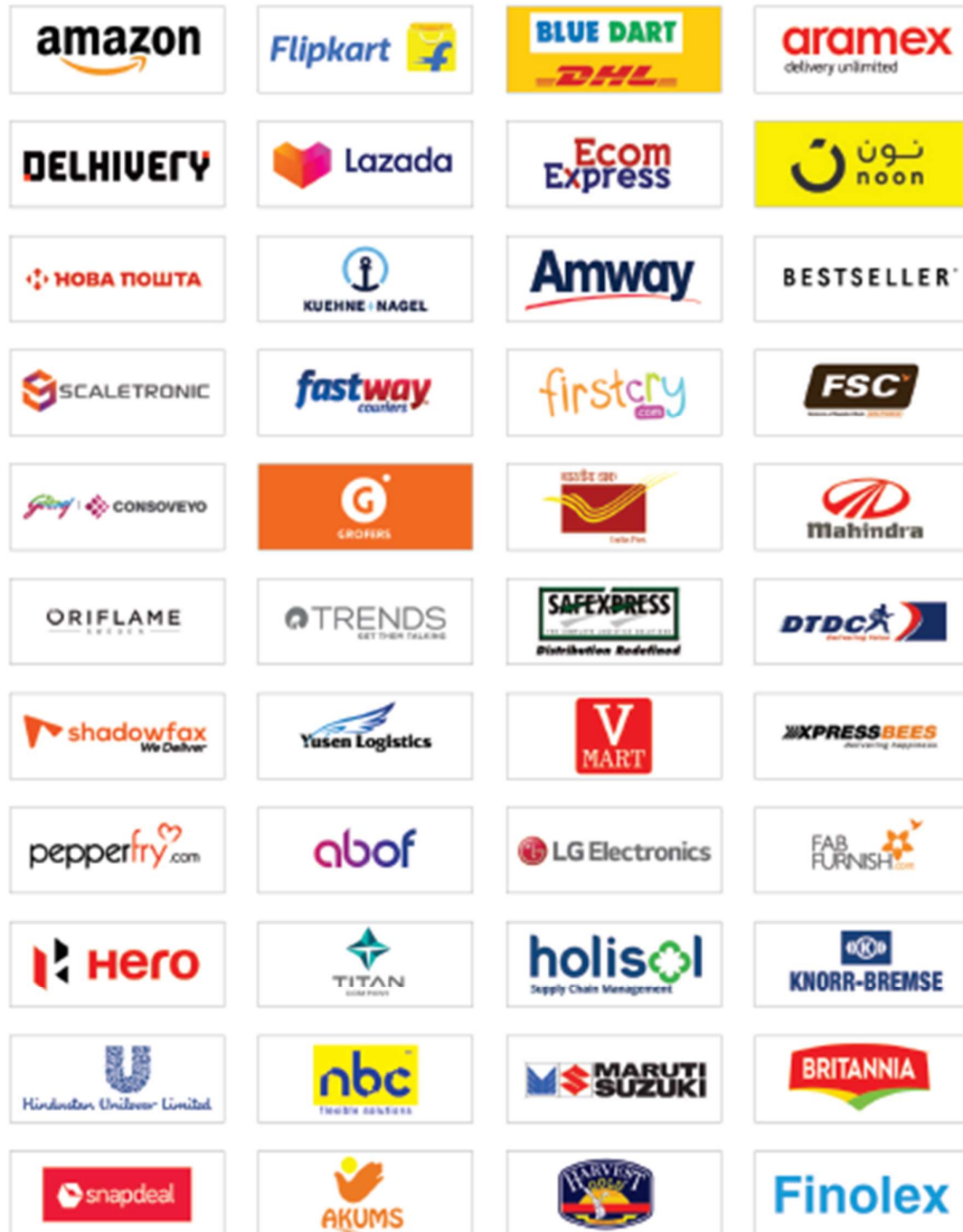
Falconautotech's unique and proprietary Automation Systems Management Engine: **AMS** is a one of its kind software platform available in the market today that opens up tremendous avenues for:

- Predicting and avoiding unplanned down times.
- Increased visibility and control into health, maintenance and functioning of the equipment.
- Efficient Spares Management System.
- Seamless Knowledge transfer for system maintenance and troubleshooting.



SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.



INDUSTRY SOLUTIONS

As a leading player in Intra Logistics Automation Space, Falcon Autotech continuously strives to improve the operational efficiency and accuracy for its clients through its domain knowledge consultancy and experience in addition to its wide range of Products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in Select Industries that lie in the Forte of Falcon Autotech and deliver world class projects in return.

FOCUS INDUSTRIES



E-COMMERCE RETAIL



COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



SPARE PARTS DISTRIBUTION



GROCERY DISTRIBUTION

FOCUS APPLICATIONS FOR FALCON AUTOTECH

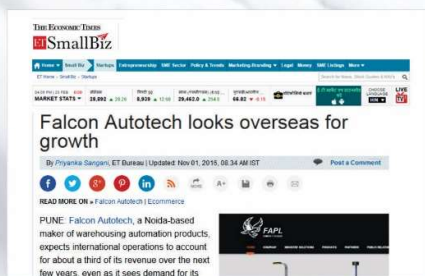
- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling

MEDIA COVERAGE

NO.1 LOGISTICS AUTOMATION COMPANY IN INDIA*



Falcon Autotech has achieved high recognition in the Intra-Logistics Automation world, "Global Logistics Automation Market" report published by Data Bridge Market Research has featured Falcon Autotech among top 20 in global list of Sortation Systems manufacturers. CNBC Awaaz, Awaaz Entrepreneur featured Falcon Autotech as fastest growing Logistics Automation company in India, Besides being featured in many important Daily Newspapers, we have also won the "Best Warehouse Automation Company" of the Year award at Express Logistics Supply Chain Conclave held in Mumbai in the Month of September 2015.



*Source: Report by Data Bridge Market Research

CLIENT TESTIMONIALS

“ Working with Falcon Autotech has been a very delightful experience for us, BLUE DART being a leader in the express logistics business, we expect the same kind of responsiveness, reliability, commitment & passion from our partners and Falcon has always delivered to their promise on all the solutions such as the Auto Sorter, Weight Dimension Labelling machine, Parcel Locker etc. All the technology interventions have been business critical for us.



Anil Khanna, Managing Director
DHL BLUE DART

“ Falcon is an organisation that has shown great agility and has helped us improve our service capabilities. Their sortation systems have helped us deal effectively with our growing volumes. We have found the Falcon team quite responsive to feedback and quick to incorporate changes into their systems that have helped build better, customized solutions for Delhivery.



Sahil Barua, CEO
DELHIVERY

“ Falcon Autotech has been efficient and proactive, particularly in meeting deadlines and their technical advancements. Falcon Autotech strike just the right balance between being a flexible, easy to deal with firm yet having all the key strengths and attributes. Every member of the team is honest, reliable, responsive, personable and very knowledgeable. I wish them good luck for all the future endeavours.



Sanjeev Saxena, Director
ECOM EXPRESS

“ I believe you guys earned it with your hard work and dedication. I believe the products you have designed demonstrate innovation and lateral thinking. Also the management has a flexible and the customer centric approach. It has been a delight to work with you and the team.



Amitava Saha
CEO, firstcry.com

- Currently possess one of the **World's largest portfolio in Sortation Technologies:** 7 In-house technologies.
- Total installed capacity of **6 Million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.

Born2Invest
Since 1998, we are helping you grow your business.

TOP STORIES ▾ **CROWDFUNDING ▾** BUSINESS ▾ FINANCE ▾ CRYPTO ▾ COMMODITIES ▾ LUXURY ▾ TOUCHPOINTS ▾ SO

Incas-SSI Schaefer merger creates leading player in Italy's intralogistics market

Although, the transportation management segment of the logistics automation industry is expected to have a higher CAGR. This is due to factors associated with the growing need to deliver goods in a timely manner and to reduce transportation costs.

Moreover, automotive will be the leading segment of the industry from 2018 to 2023, with an increasing demand for automated production systems and flexible logistics systems used in manufacturing and supply of vehicle fleets.

The MarkestandMarkets research report has identified SSI Schaefer as one of the key players in the logistics automation market worldwide during the forecast period, along with Dematic Corporation (US), Murata Machinery, Ltd. (Japan), Jungheinrich AG (Germany), and Falcon Autotech (India), among others.

This year's largest market for logistics automation is estimated to be North America, according to the research report. Regardless, Incas and SSI Schaefer's merger could help them compete in their main market.

(Featured image by [DepositPhotos](#))

0







Not secure | worldindustryinsights.com/30269/global-parcel-sortation-system-market-analysis-size-share-growth-trends-and-forecast-2019-2026-siemens-vanderlande-beumer-group



Global Parcel Sortation System Market Analysis, Size, Share, Growth, Trends, and Forecast 2019-2026: Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives

BY ANN.CASTRO ON AUGUST 12, 2019

The global "[Parcel Sortation System market](#)" research report is a pervasive research report that touches the most vital parts of the Parcel Sortation System market that is necessary to be grasped by a professional or even a layman. The market research report enlightens one regarding few of the important aspects such as an overview of the Parcel Sortation System product, the growth factors enhancing or hampering its development, application in the various fields, major dominating companies, genuine facts, economic situation, and geographical analysis. The research report endows the data concerning the dynamics that propel the growth as well as the supply and demand chain of the product on a global basis.

Click here to access the report: www.reportsbuzz.com/request-for-sample.html?repid=50822

The international and global stand of the Parcel Sortation System market is also briefly mentioned in the research report based on the performed statistical and thorough market analysis. The information mentioned in the research report gives a qualitative and quantitative view of the overall market. The statistical analysis of the market helps analyze the supply, demand, production, maintenance, and storage costs of the product. The data regarding some of the prevalent players Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives, Dematic, Interroll, Muratec, Invata Intralogistics, Viastore, Okura Yusoki, GBI Intralogistics, Equinox Mhe, Pitney Bowes, Falcon Autotech, OCM SRL, Solystic, Bowe Systec, Intralox are also detailed in the current case study.



4. Reference projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Piece Picking and Handling, Case Picking and Handling & Parcel Handling
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

4.1 Project 1- (Ecommerce Client)

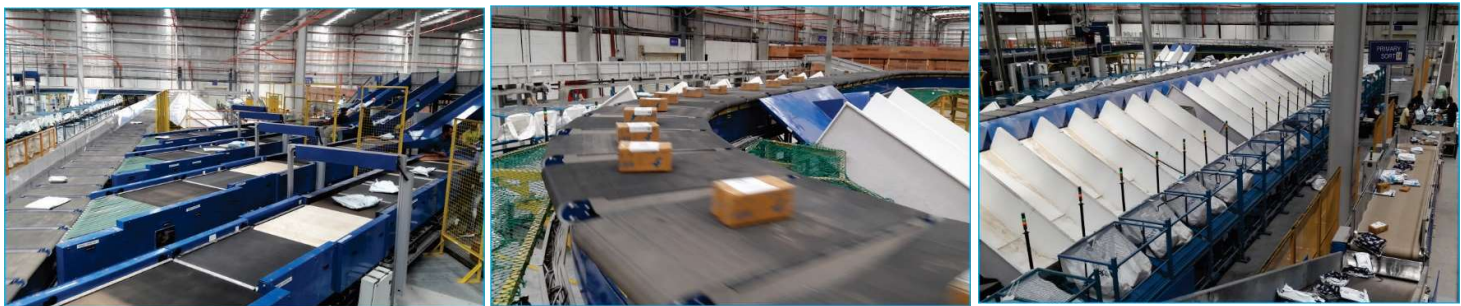
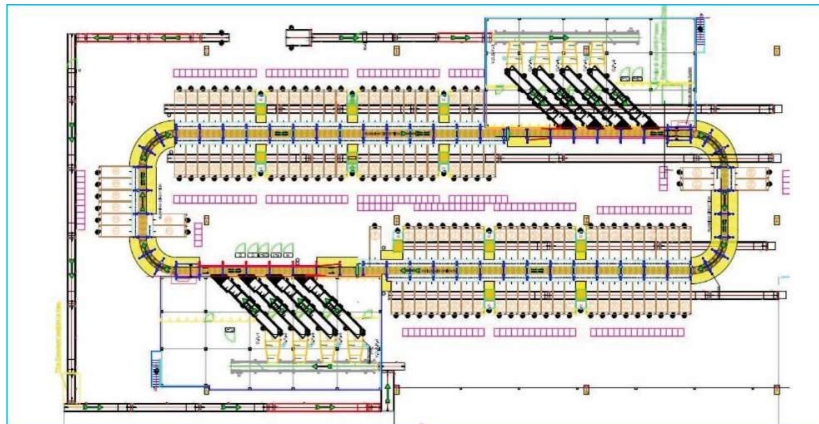
Use Case - Destination Sorting of Packed Parcels

Solution Specifications –

- Throughput: 16,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



Layout & Pictures_ Project 1

4.2 Project 2- (Client – Courier Express Parcel)

Use Case - Courier Sorting of Parcels

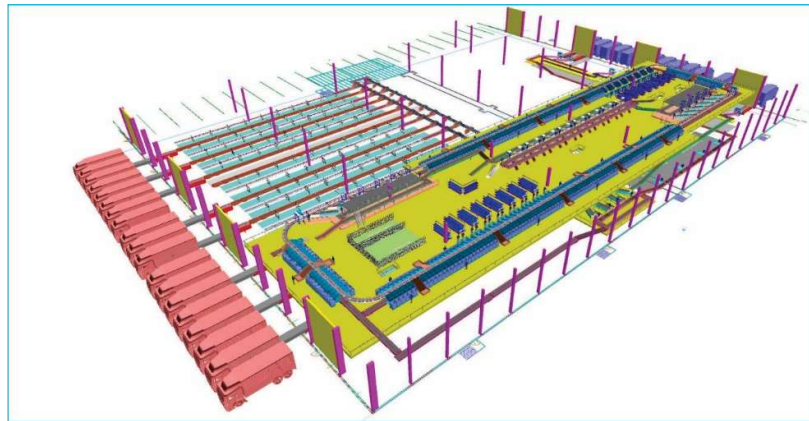
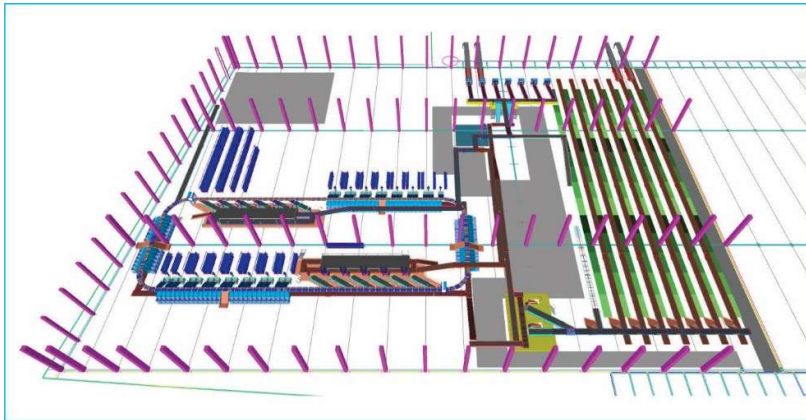
Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 394 Direct, 30 Secondary, 630 PTLs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Pre-Sorting
- Telescopic Belt Conveyors
- Bag Sorters
- PTL System

Layout -



Layout & Pictures_ Project 2

4.3 Project 3- (Client – E-Commerce)

Use Case – Destination Sorting of Packed Parcels

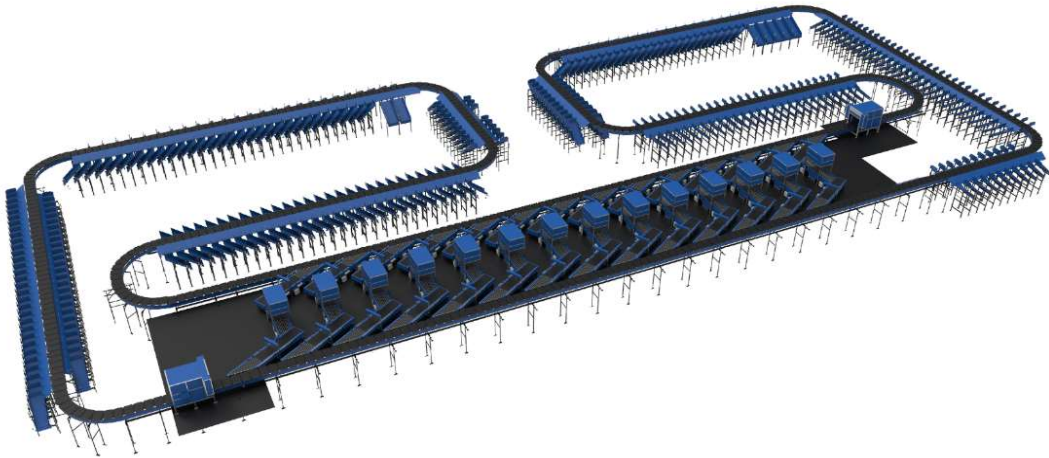
Solution Specifications –

- Throughput: 26,000 PPH
- End Destinations: 410
- Building Size: 500,000 Sq Ft

Key Technology Modules –

- Bulk Flow Sorting
- Automatic Pre-Sorting
- Loop Cross Belt Sorter with cell size of 900 x 500 mm
- Bag Conveyors

Layout -



Layout _ Project 3

4.4 Project 4- (Client – Indian Retailer)

Use Case – Store wise Sorting of Apparels

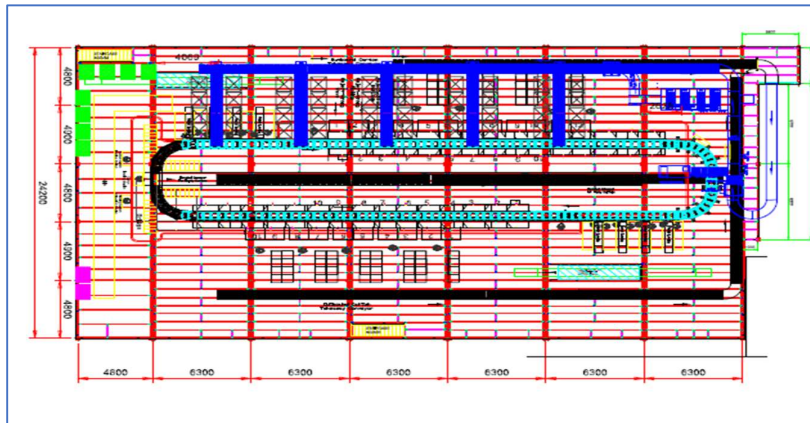
Solution Specifications –

- Throughput: 20,000 PPH
- End Destinations: 1750 using PTLs
- Building Size: 700,000 Sq. Ft

Key Technology Modules –

- Loop Cross Belt Sorter with a cell size of 600mm x 330 mm
- Belt Conveyors
- Case Sorters
- Telescopic Belt Conveyors
- Packing Machines
- Vertical Lifters

Layout and Site Pictures –



Layout & Pictures_ Project 4

5. Handled Parcel Spectrum

5.1 Parcel size loadable on the sorter

Gaylords	Max Length	mm	600
	Max Width	mm	600
	Max Height	mm	400
	Max Weight	Kg	50

Sacks	Min length	mm	900
	Min Width	mm	600
	Min Height	mm	N/A
	Min Weight	Kg	32

5.2 Parcels to be loaded on Sorter shall have the following characteristics:

1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behavior of parcel cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

5.3 Parcel not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items.
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured.
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.

13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

5.4 Pictures of Sortable Parcels

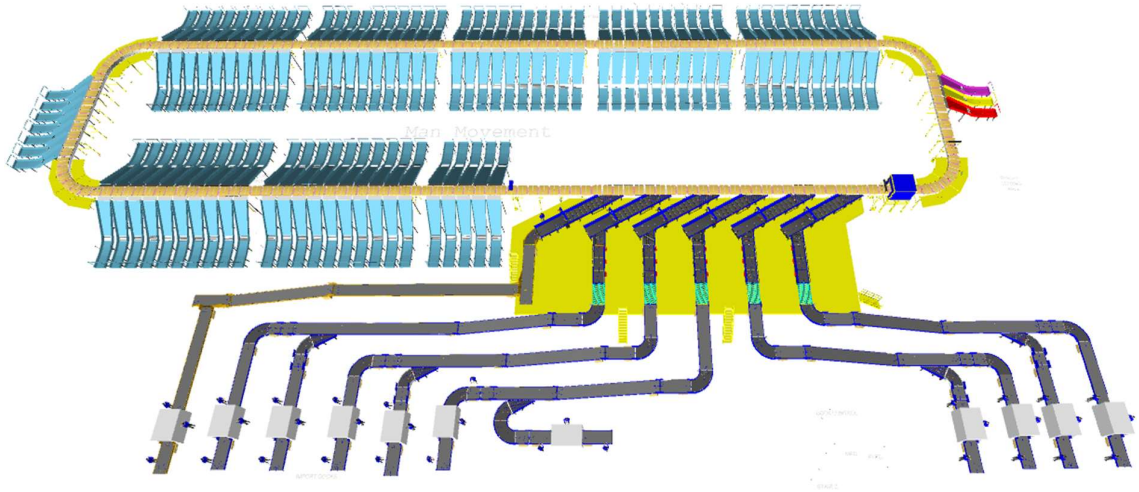
PARCEL TYPE		DESCRIPTION
	Strongly Strapped	A parcel banded with strapping material.
	Pre Manufactured Plastic Courier Pack <A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre Manufactured Plastic Courier Pack > A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed.
	Other Plastic Covered Item	A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item	A cardboard parcel coated in a packing paper.
	Cloth Covered Item	Often Import parcels that are of irregular shape.
	Wine Box	A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box	A cardboard parcel is often the norm.
	Totes	A large plastic, lidded box.
	Flats	A parcel that often contains documents.
	Poly Bags	It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.

6. Proposed System Description

6.1 Objective

The purpose of this proposal is the design, manufacturing, installation, commissioning, testing and acceptance testing of a complete Sortation system to handle Parcels.

The proposed solution also enables the connection of the Parcels processing system to the steering, as per the interface defined by Emirates SkyCargo.



6.2 Summary of the System

1. Dual Belt XL cross belt Sorters to be Installed on the ground.
2. Shipments and bags fed to the infeed conveyors. These shipments are then pass through X-Ray machines. Each infeed line is installed with 2 X-Ray of 1000 PPH each.
3. From Infeed conveyors the shipments are conveyed to the Induction Line.
4. Operators orient and places the shipments on the Induct lines equipped with sensors
5. Feedlines induct the parcels on the Sorter automatically onto to the Cross-Belt Sorter Loop which then sorts the parcels into respective chute using data from Emirates SkyCargo 's Sorting decision system.
6. As per Emirate logic and parcel condition the sorted shipments are discharged into Collection Chutes, Rejection Chute, Hold shipment chute & Relabeling Chute.
7. There is a separate feedline for Bags. Bags are fed to infeed where it will pass through X- ray machines.
8. Safety devices are considered as per Falcon Standards and referenced from previous projects.

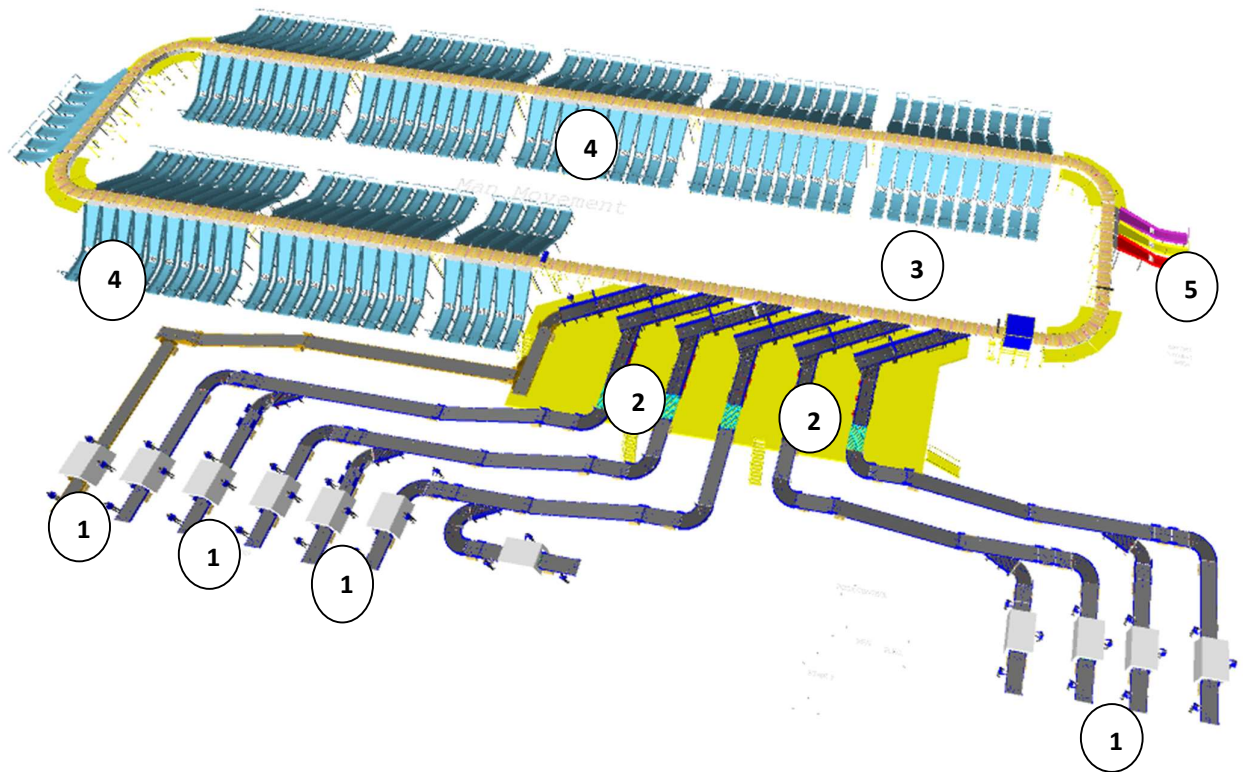
9. Control and IT Systems to suit the requirement and as per Emirates SkyCargo 's specification.

6.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput
2. Good Parcels conveying quality for the entire range of conveyable Parcels with a sorting conveyor speed up to 2.1 m/s.
3. Low occupancy of floor space in the building
4. Narrow discharge centers for increased number of splits in limited space
5. Scalable with increasing Business Volumes.
6. FALCON's CBS can adapt with changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear

7. Layout View



Legend

- 1 – Infeed Line
- 2 – Feedlines
- 3 – Main Loop
- 4 – Collection Chutes
- 5 – Irregular conveyor

8. Proposed System Capacity_ Calculations

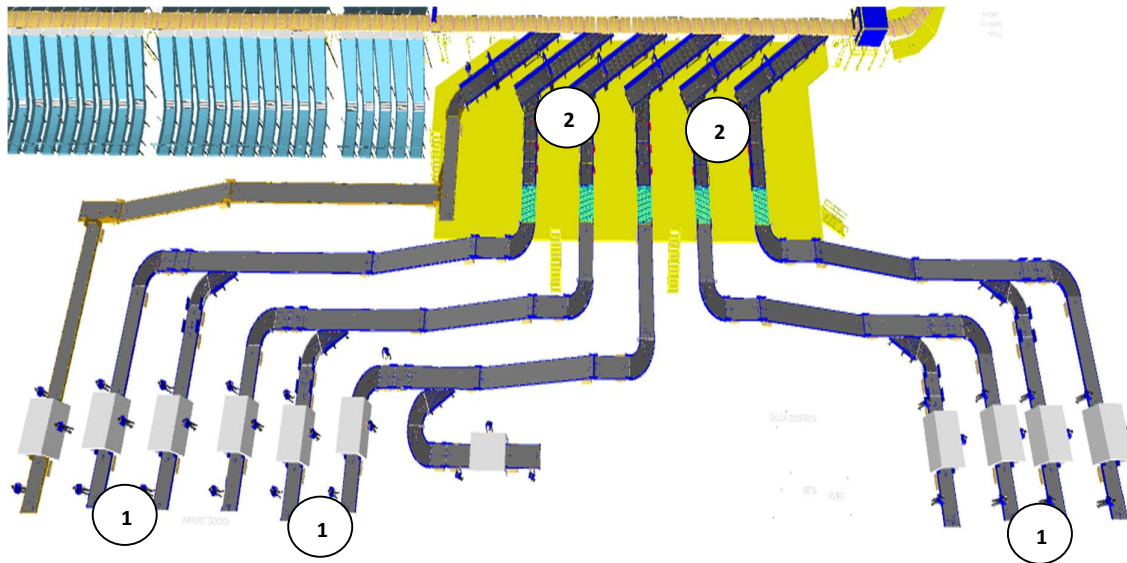
8.1 Sorter System Capacity

The following table shows the Throughput Calculation for the Sorter

Specifications	Value
Sorter Type	Loop CBS XL
Design Throughput	9000
Carriers per hours	12000
Speed- m/s	2.1 m/s

System description

8.2 Infeed System



Legend

- 1- Infeed Conveyor
- 2- Feedlines

1. Handheld Barcode Scanners

Each Feed Line will be equipped with a Handheld wireless 2D Barcode scanner This scanner works as a backup for Auto scanner as well to ensure higher system redundancy.



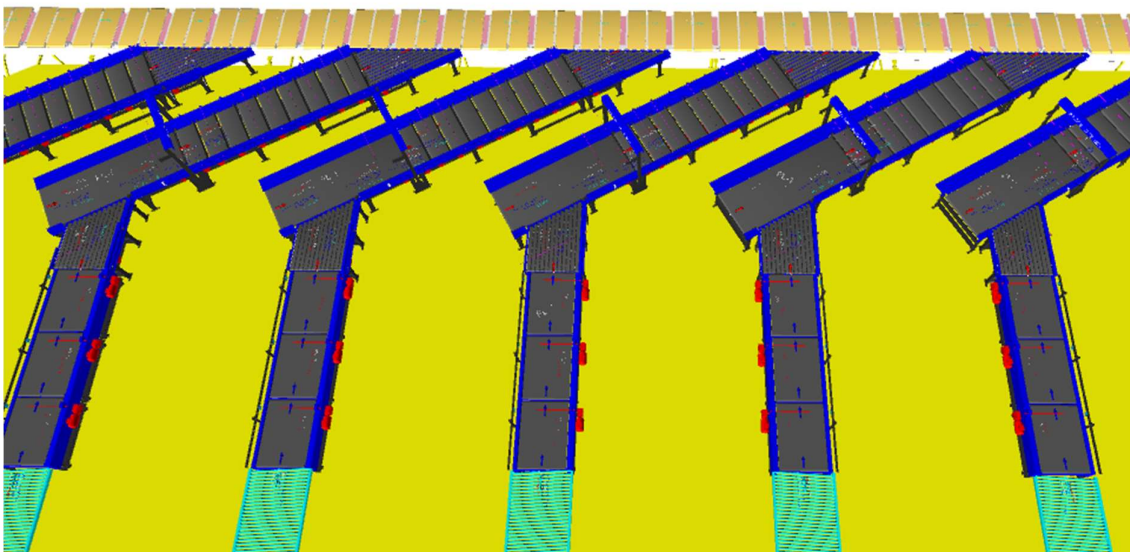
2. Siemen's HMI

Provision of a HMI at Induct Stations which enables visualization and control of Process.



8.3 Feedlines

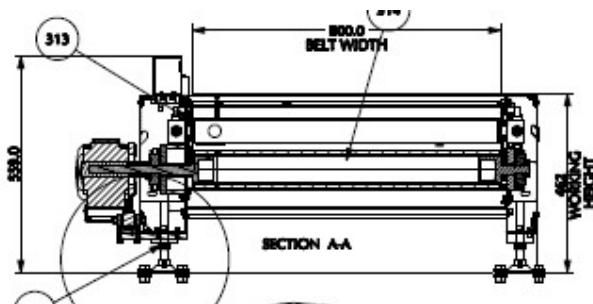
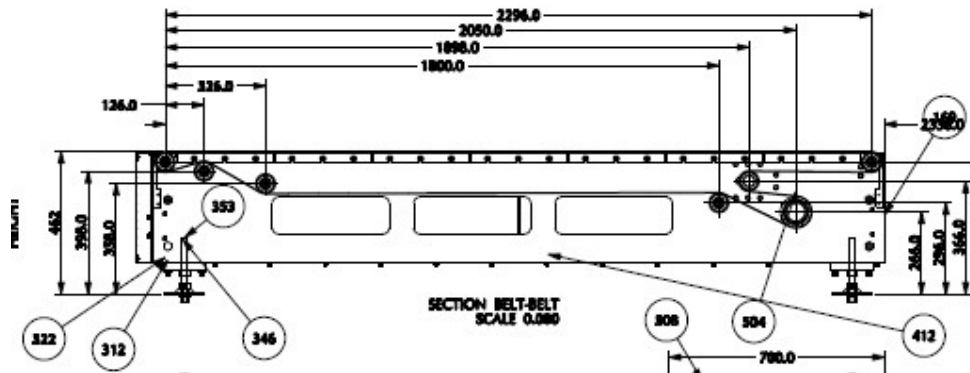
The System has total 06 Automatic Feedlines with Designed IPP of 2000 PPH.



1. Each Feedline has Four Conveyor Modules- 9 Short Buffer conveyor, 1 Receiving conveyor and 1 angle merge conveyor.
2. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Parcel on the conveyor and prepare the Cross-Belt Cell for receiving the parcel with high precision.
3. Over the feedline, parcels are evenly spaced and smoothly transferred to the main Loop.
4. Size Gauge will be provided at Induct Stations to help Operators to determine the Oversized and Undersized Parcels.

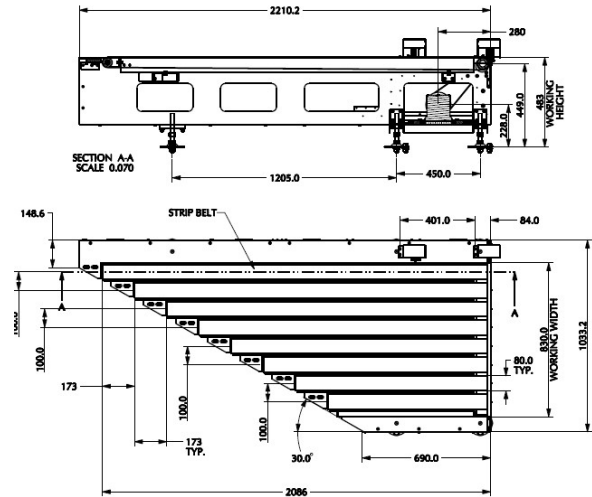
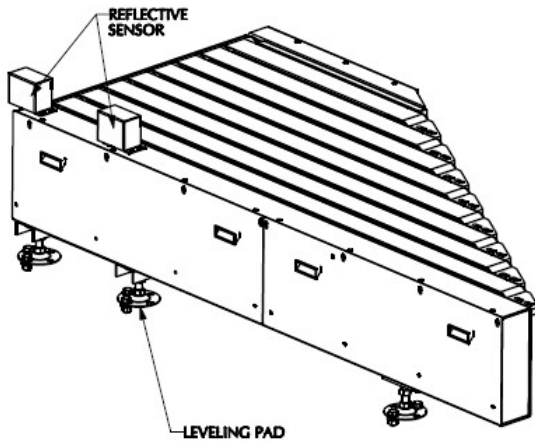
8.3.1 Short Buffer Conveyors

High-speed conveyor, used to create appropriate space between parcels/boxes and synchronize the infeed into the Sorter.

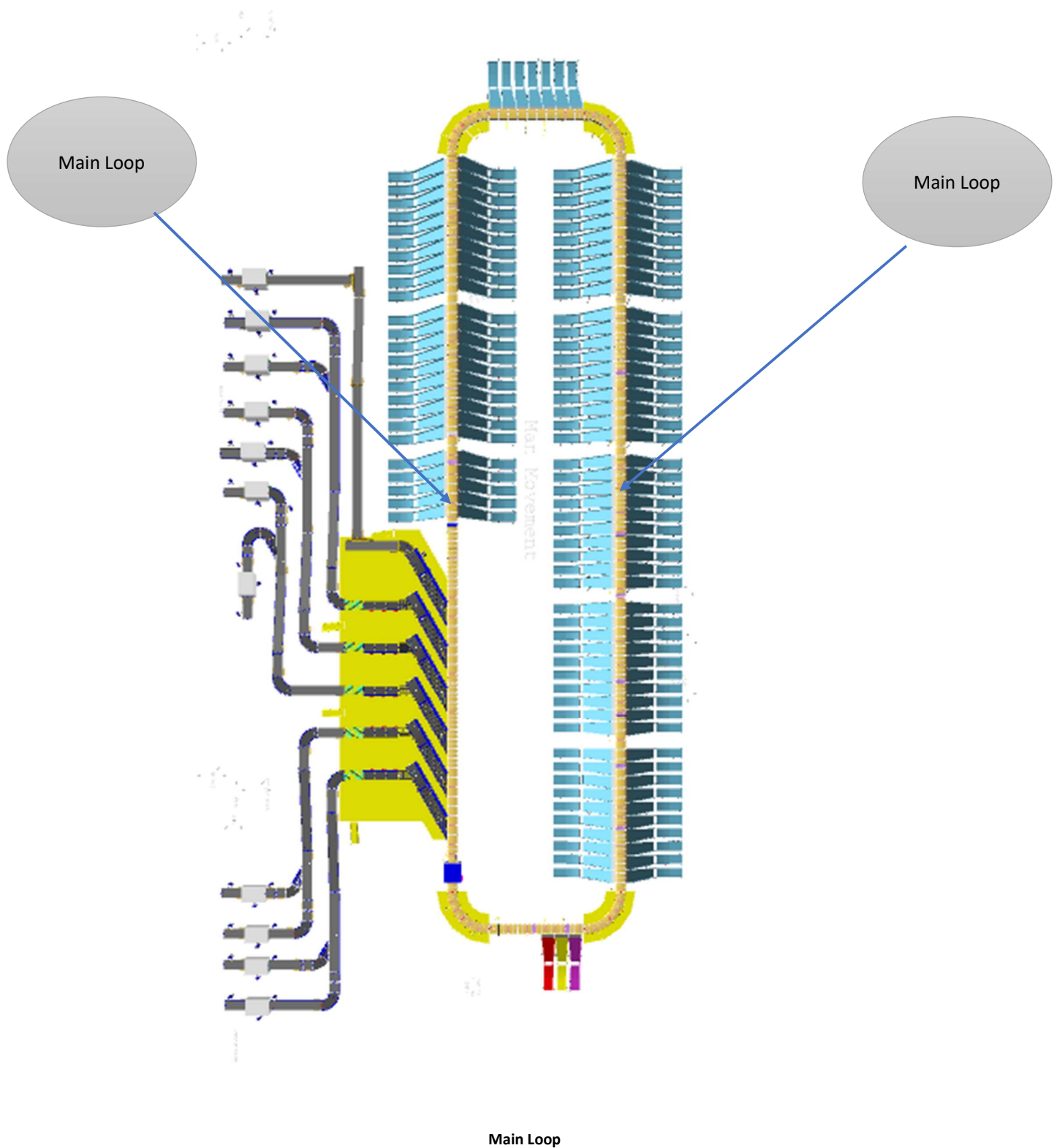


8.3.2 Intelligent Merge Conveyor.

This is 30° triangular Dual Belt high-speed conveyor used for inducing parcel/boxes directly on to the sorter. The Belts are Strip Belts for smooth parcel movement.



8.4 Main Loop



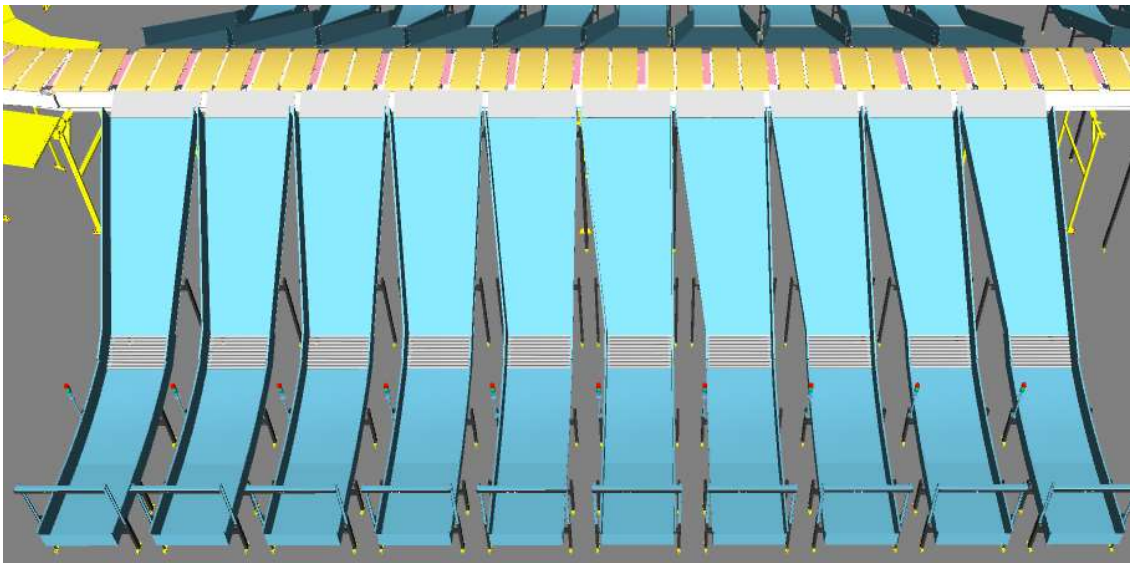
1. The proposed Double Deck CBS Sorter will be installed on the Mezzanine. The Sorter is running at 3400 mm from Ground level.

2. We have provided Dual Belt Carriers with 1175 mm pitch and Belt Size of 1175 x 800mm.
3. As Parcels pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the shipment reaches respective chute, carrier carrying the shipment for the chute, will actuate and drop the shipment in the assigned chute.

8.5 Output Chutes

1. Collection Type Chutes.

The sorted shipments will travel along these L-Shaped Sliding part and get accumulated in the Collection Part.



Chute IOs	Description
Single Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation.
Four Color Tower Lamp	To indicate the operator action on a chute
Push Button	To Start / Stop Sorting Operations

9. Proposed System Technical Details

9.1 Mechanical equipment

Qty.	Description	Value
1	Infeed System Powered Belt Conveyor – 11 Modules; 26.4 m	~1200 mm width ~1200 mm width
2	Falcon Feedlines (6) Belt Width System Height Each Consists of 9 x Short Buffer Conveyors 1 x Intelligent Merge Conveyor	~ 800 mm ~ 800 mm from Mezz Top
1	Falcon CBS Sorter Sorter Height from Ground Sorter Loop Length Belt Carriers Sorter Speed Sorter Drive Carrier Pitch Including: Standard Sorter Supports 2 X ICR Scan Tunnel 2 X Product Positioning Systems 2 X Product Centring System 2 X ECDS System- Camera Based 1 X Carrier Positioning System 1 X Speed Monitoring & Auto Correction System	~3400 mm ~240 m 149 Nos. Up to 2.1 m/s LIM 1175 mm
1	Destinations/Chute Collection Chutes Rejection Chutes Hold Shipment Chutes Relabelling Chutes Chute full Sensors Push Buttons	157 Nos. 01 Nos. 01 Nos. 01 Nos. As per design As per design
1	Steel Works Platform for Feedlines Maintenance Platforms for ICRs	Included

1	Standard FK Safety Equipment	Included
---	-------------------------------------	----------

9.2 Electrical Equipment

1	Consists of Main Power Distribution panel Main Control Panel Feedline Control Panels Sorter Drive Panels Network Switches Field Cabling	Included
---	--	----------

9.3 Control System

1	Consists of Siemen's PLC based Control System with SCADA Industrial Switch	1 Nos. As per requirement
---	---	------------------------------

10. Description of Components of Equipment

Elements of the sorting system

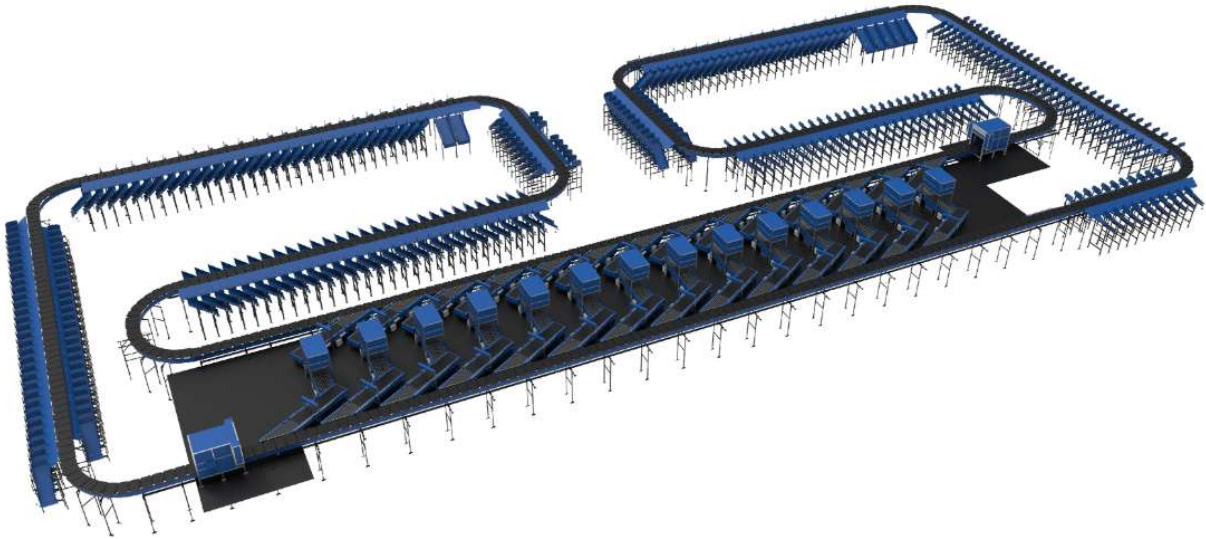
1. Cross Belt Sorter
2. Scanning & Sensing on Sorter
3. Conveyor System
4. Steel Works- Mezzanine & Staircases

10.1 Cross Belt Sorter -

Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels.

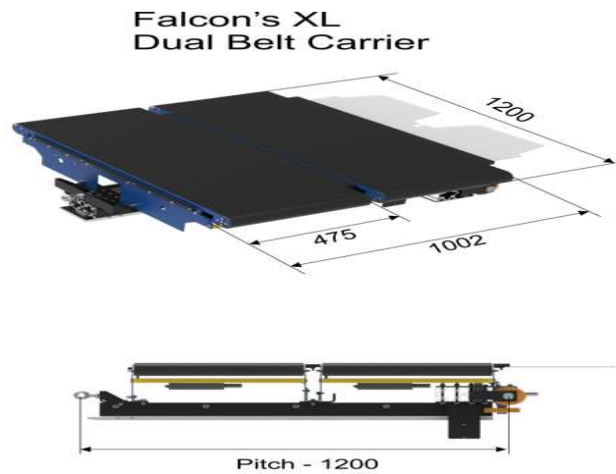
Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



10.1.1 CBS Carrier

We have provided CB XL Carriers with 1200 mm pitch.

Drawings of CBS Carrier



**Different Views of
Carrier**

CBS



Falcon Autotech Dual Belt
CBS Sorter

10.1.2 Servo Roller

High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.



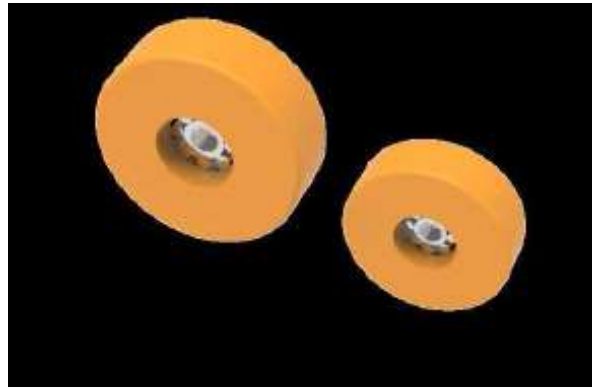
10.1.3 Chassis

Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminium which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.



10.1.4 Carrier Wheels

Carrier wheels that are thoroughly tested and proved for long life cycle



10.1.5 Non-contact based linear motor drive

Extremely **High Energy Efficient** Linear Induction Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed



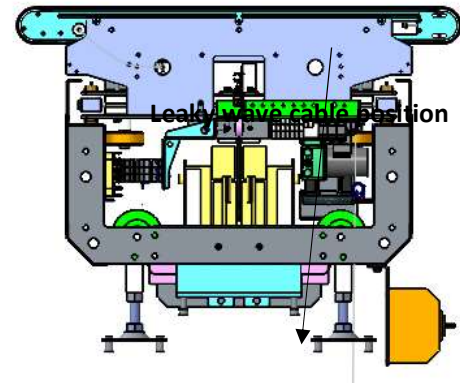
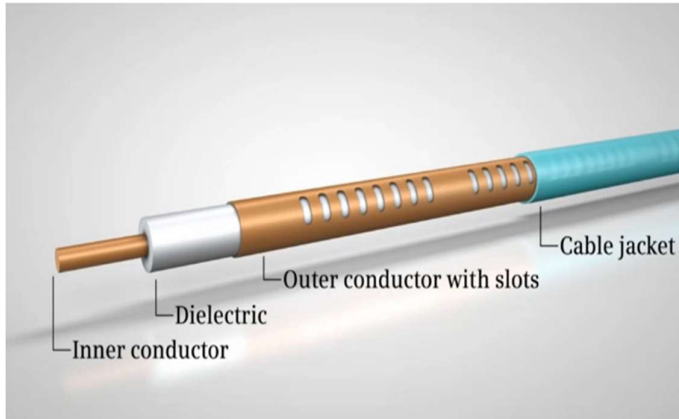
10.1.6 Power Transmission

Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.



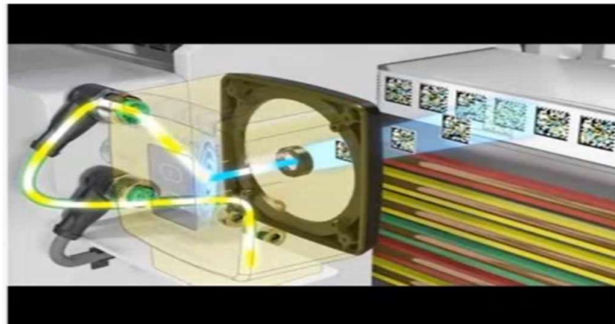
10.1.7 Data Transmission

The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.



10.1.8 Carriers positioning System.

Positioning system determines the exact location of each carrier in the loop at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter loop, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



10.1.9 Technical specifications of Sorter

Items	Make
Sorter Carrier Type	Loop CBS Dual Belt
Sorter Speed in m/s	Up to 2.1 m/s
Sorter Loop Length- Lower Deck-	~240 m
Length of the Sorter	Layout Attached
Width of the Sorter including Chutes	Layout Attached
Sorter Actuation Technology	Electric
Noise Level	<70 Db
Chutes	
Chutes L Type collection + Rejection	40 + 4
Height of Sorter-Lower Deck -	~34000 mm
Track Material & Type of Design to move the Carrier trolleys	~Steel Track
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch - mm	1175
No. of Carriers	149
Drives	
Motor/Drive type	Linear Induction Motor
Power Consumption	Will be specified during DAP
Induction Mechanism to the Sorter	
Feedlines (Automatic)	6 Nos
No. of Barcode Scanners (ICR type)	01 Nos
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Empty Carrier Detection System (Camera Based)	01 Nos
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Refer Layout

10.2 Scanning & Sensing on Main Sorter Loop

1. Automatic Barcode Scanning & Dimensioning System

Top Sided Barcode Scanner to scan parcel 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time.

Handheld Barcode Scanners are provided to run the machine on Semi- Automatic Mode, if required



2. Parcel Positioning System.

Falcon Cross Belt Sorter is equipped with Multiple parcel positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Parcel on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).



3. Empty Carrier Detection System

This system detects the empty carriers on the sorter and prepares the Feedlines to load the parcels on the empty carriers. (Camera based)



10.3 Conveyor System.

a. Belt Conveyor

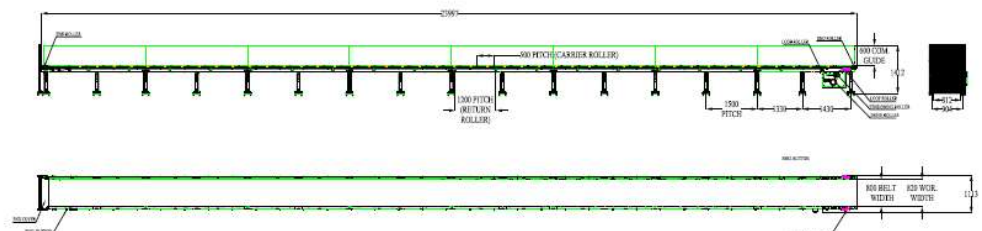
This paragraph presents the main characteristics concerning straight or inclined conveyors.

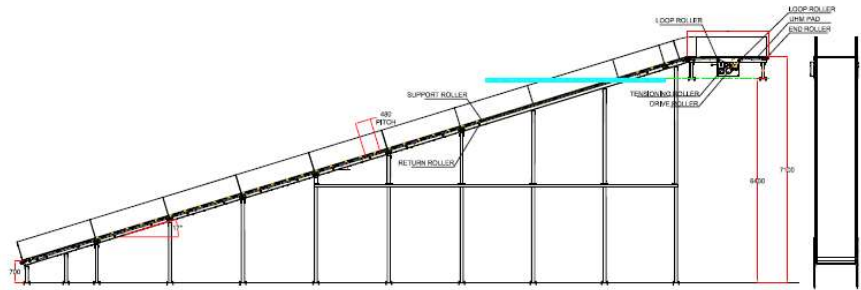
Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.

Some Silent Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.

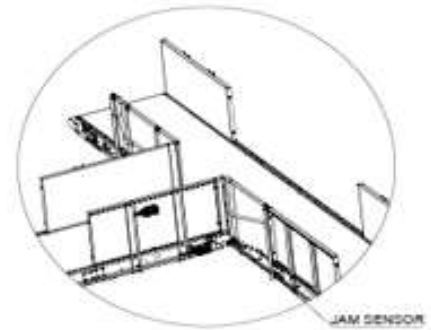
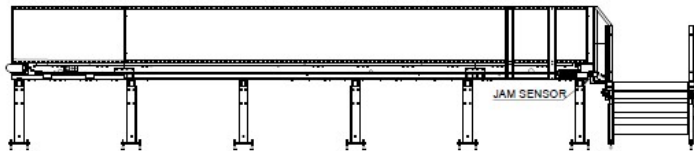




PVC Belt Horizontal & Inclined Conveyors

b. Waterfalls

A 100 mm of drop has been provided at 90-degree transfers.



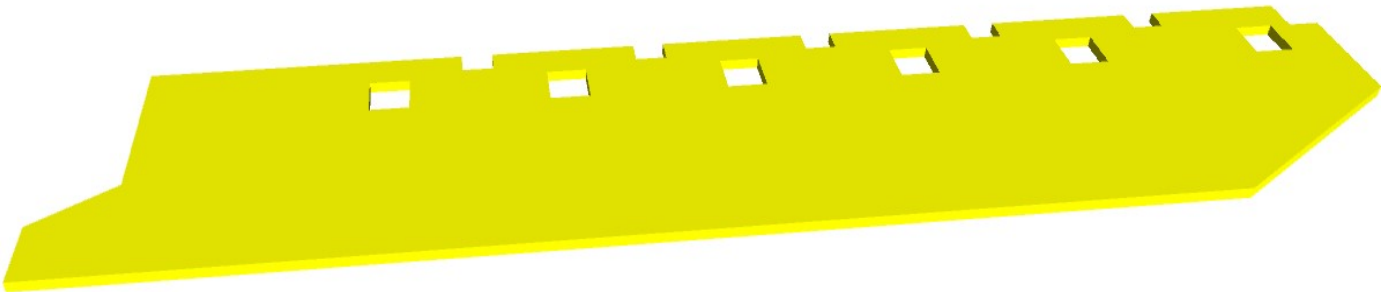
c. Conveyor BOQ

Row Labels	Sum of Set	Sum of Total Length (m)	Sum of Selling Price
Powered Belt Conveyor	53	219.9755	18524265
Scanning Conveyor	11	26.4	2750000
FAL_BT60°_W1200	6	0	4950000
Modular Conveyor	5	50.88	4523180
Belt Merge_1200	5	0	1825000
FAL_BT90°_W1200	11	0	9075000
FAL_RC_Aligning Conveyor_W1200	5	11.75	1410000
Grand Total	96	309.0055	43057445

10.4 Steel Works – Mezzanine, Maintenance Platforms & Staircases-

a. Operator Platform -

1. Required at Induct Zones



Total Area: Operator Platform

Total Induct Platform Area - Approx. 496 Sqm

b. Staircases

Staircase Qty- 04 Nos.



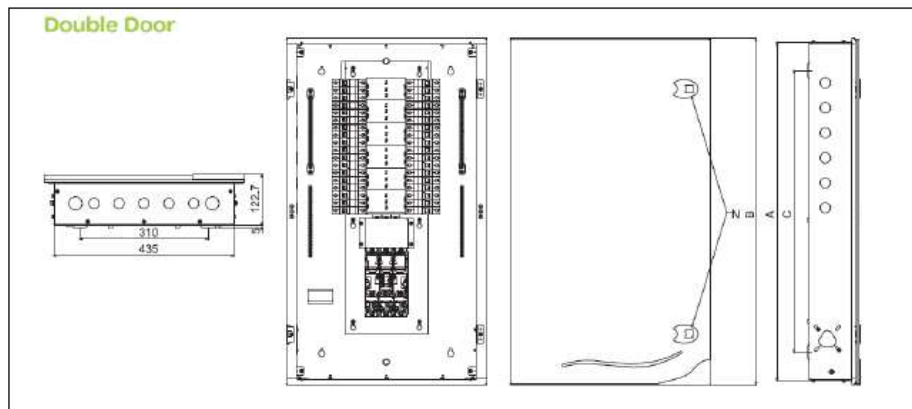
11. Electrical System

Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets.

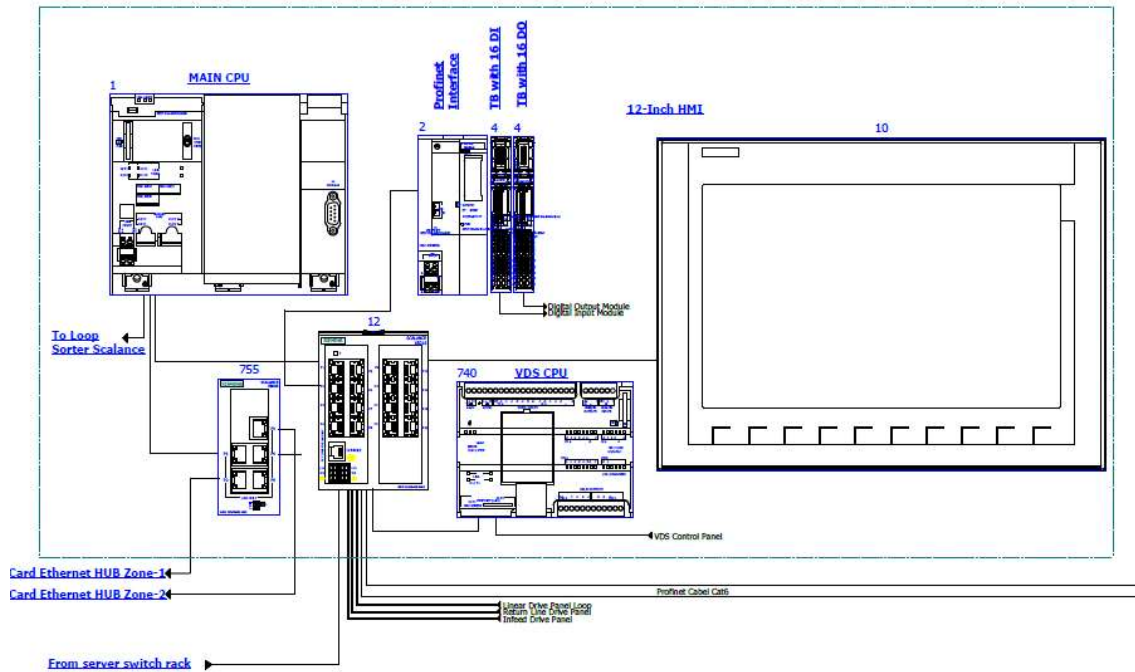
PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

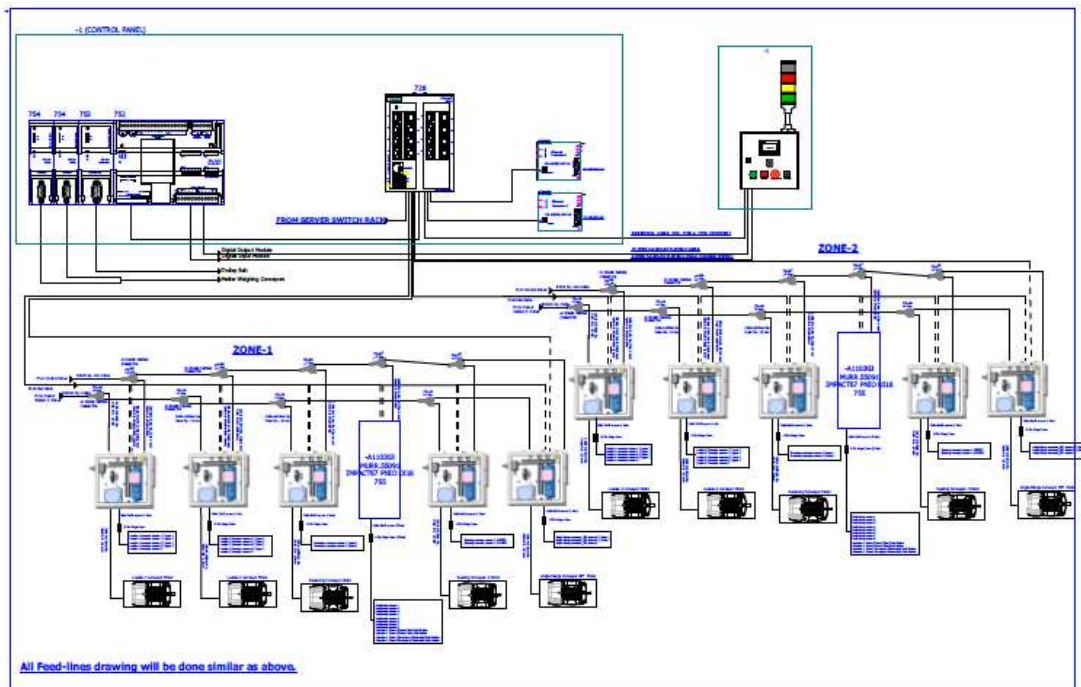
a. Reference Picture of Power Distribution Panel



b. Main Control Panel (Reference)



c. Induct Stations Control Panel (Reference)



Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter.

The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.

All motors will have appropriate IP ratings

Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

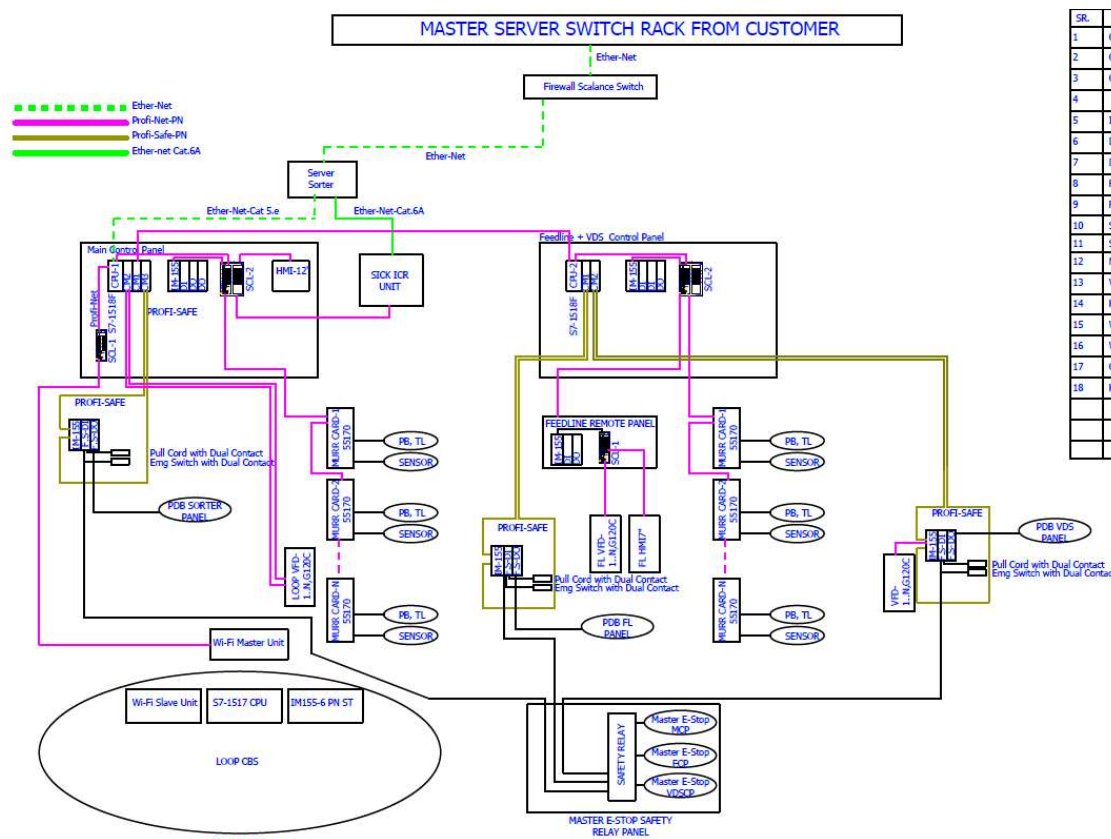
Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform. The entire system will be logically divided into Zones (Sorter/ Feed Line/ Loop), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet.

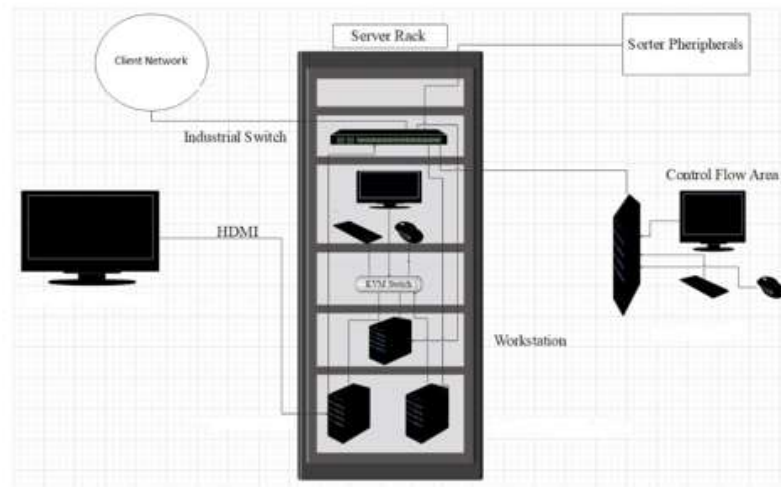
Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

12. Controls Architecture

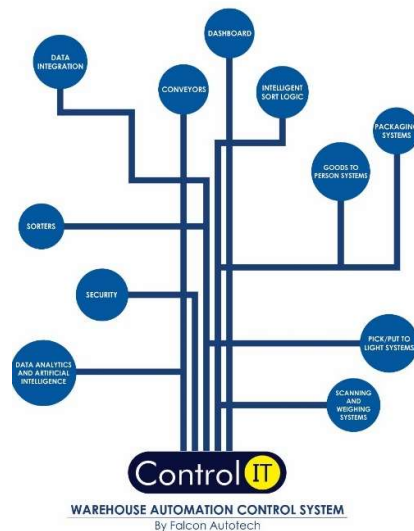


13. IT Architecture (Ref.)



Note – Main & DLIC Server System & Server Rack in Emirates SkyCargo 's Scope

14. Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

15. DLIC (Deep Learning Image Classification)

Artificial Intelligence built where the software is trained to automatically read and classify the captured images of products and process them as per defined protocols

The **image-based system “Single Parcel Verification”** can detect single or multiple Parcels using the images of an ICR890 camera. This kind of singulation verification prevents Mis-sorts

Benefits -

- Reduce delays and costs by avoiding double occupancy of cells
- Reduces Mis-Sorts
- Enables smarter operations leading to a leaner operational style

Detection Rate Single > 99%

Detection Rate Multi > 95%

One Year Subscription; Need to be renewed every year

DLIC Server in Emirates SkyCargo 's Scope

Operational Requirement -

Conveyor surface	Homogeneously dark	
Objects (both)	Obviously seen in the image (at least 20% of both object are seen)	
Objects with patterns	Causes lower detection rates	
Object remission	15 ... 200 % (others cause lower detection rate)	
Image distortions	Not permitted	
Banded objects	Not permitted	
Object restrictions	Trays/totes, transparent consolidated bags or shrink wrapped bundles are not permitted	
Object consistency	Destroyed objects or objects with hanging parts are not permitted	
Guarding	Guards visible in image or reflected objects no allowed	

16. Key Components Make

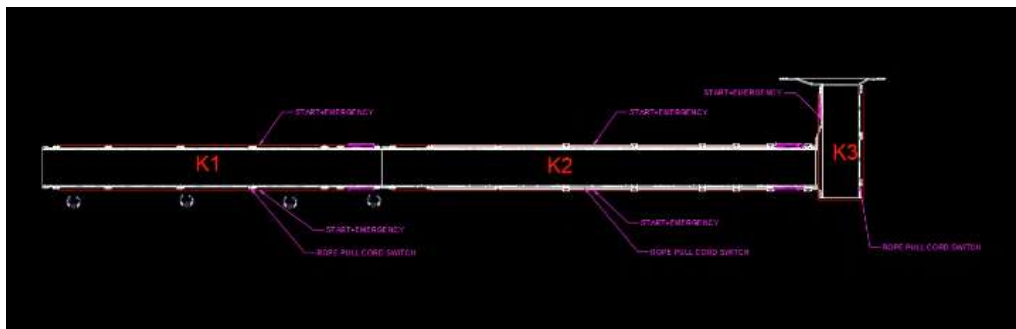
Items	Make
Belts	Forbo/ Ammareal /Habasit
Rollers	Falcon
Cross Belt Carriers	Falcon
Linear Induction Motors	Falcon
Feed Line Motors	Induction Motors – SEW/ Lenze
Barcode Readers	SICK / Cognex
Wireless Barcode Scanner	Honeywell/Motorola
Load Cells	MT
Volume Scanners	SICK
Encoders	SICK
Sensors	Sick/Leuze/ P&F
PLC	Siemens's
Control Panels	Rittal/ BCH
MDR Rollers	Falcon/ Pulse/ Interrol
Power Transmission Systems	Vahle

17. Principle of Safety

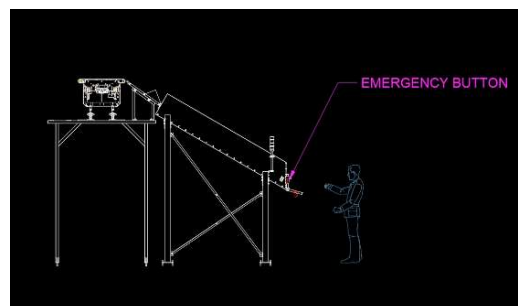
1. E-Stops



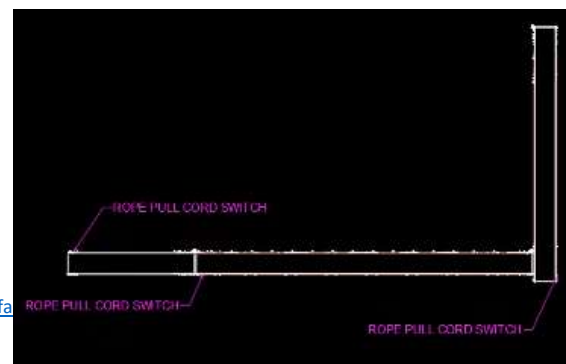
a. At Every Conveyor Module – Both sides



b. VDS Chutes



2. Pull Cords Switch- Required for Infeed & Takeout Conveyors



3. Fencing

- a. Between Inducts
- b. Between Inducts and Sorter
- c. Gates of VDS Platform
- d. Gates of Maintenance Platform
- e. Gates of Mezzanine



18. Additional Scope-

1. Provision of Fans at Inducts and Chutes for heat dissipation
One at each Induct and One at every 10 chutes
2. Provision of Curtains to be considered in front of VDS to slow down the dropping shipments

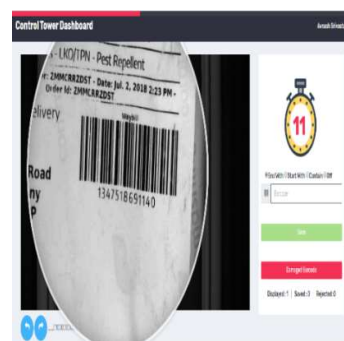
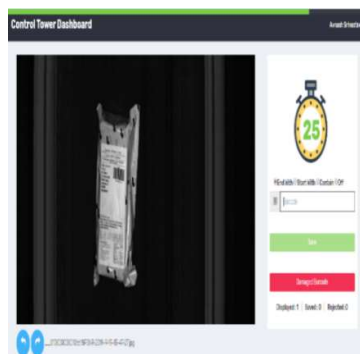


3. Safety PLC – Siemen's 1518 Series
4. Siemen's Safety Switch for SCADA

5. Safety Barricading at ground to be considered. (Safety fencing & Column Guards). Price to be agreed on per m basis during DAP; INR 5000.00 per m for fencing and INR 2000.00 for every Skid plate
6. Stress Free Mats at Inducts
7. Electrical Insulation Mats at PLC Panels
8. Estops at CBS Loop
9. FRLS Cables- LAPP/ Polycab/ KEI Make
10. Side Guards for Bag Takeout Conveyors
11. Power Saving Sensors for Conveyors
12. Electronic Barricading at Inducts as done in FRK CBS

19. USPs of Falcon's Cross Belt Sorter

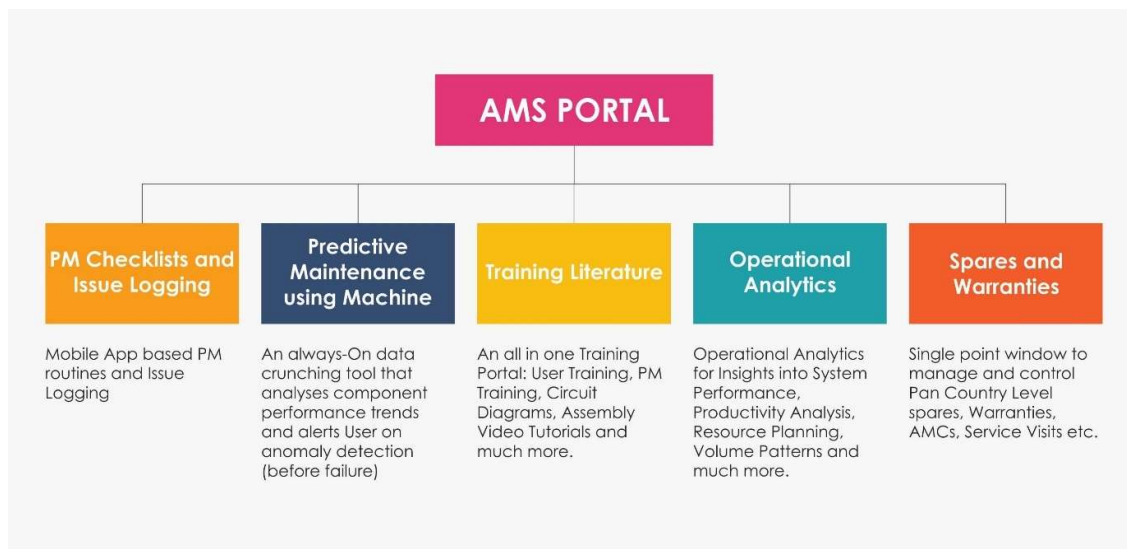
Real Time Video Coding – this new innovative feature of Control IT software directly focusses on reducing the rejections to approximately zero which arises out of barcode unreadability while the sorter is running. This allows a remote-Control Tower to Decode images from a PC and relay the decoded value back to the sorter in real time and reduce Barcode read rate related rejections drastically. This is a flagship feature in Falcon's Cross Belt Sorters.



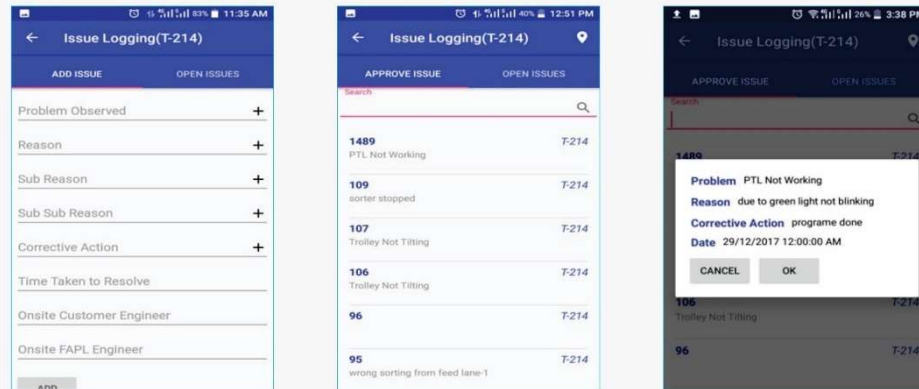
- Falcon Autotech's Cross Belt Carrier chassis is made up of lightweight Aluminium which makes them light yet sturdy. This reduced weight leads to Potential "Power Savings" over a considerable period of usage.

- Carrier wheels with large diameter are used which reduces number of revolutions for the same Track Length, thereby reducing wear and tear
- CBS Carriers can discharge a parcel anywhere along the track of the sorter, thereby ensuring 100% flexibility in Chute positioning even after the system installation. This also means that sorter can discharge products on turns, thereby increasing the chute density and reducing the overall system footprint as well as cost'
- Variable Electric Discharge Acceleration depending upon the product profile.
- Contact based Linear Motor Drive options for-
 - 1.) Low noise levels (<70 Db) meaning a healthy work environment for the work force.
 - 2.) Reduced mechanical break downs leading to high System overall availability (>98.5%)
 - 3.) Extremely low maintenance requirements due to no wear element
 - 4.) Back-up Drive units for In-built system redundancy

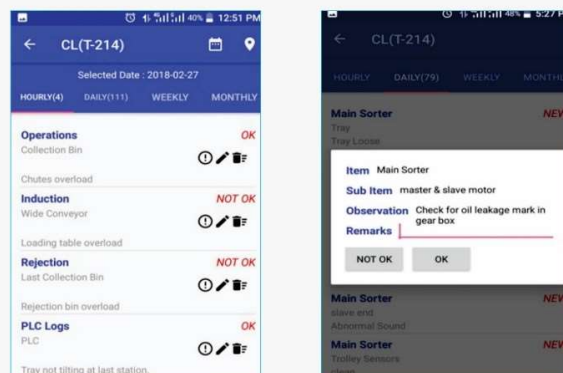
20. Automation Management System (AMS)



a. Issue Logging



b. Preventive Maintenance Check List



c. Predictive Maintenance

Notifications(T-214)	Notifications(T-214)
Sorter 27/02/2018 12:03:14 PM Packet between 2 trays after Secondary Repeated 10 Times, From 120 Seconds. Sorter 27/02/2018 12:02:13 PM Packet between 2 trays after Secondary Repeated 10 Times, From 120 Seconds. Sorter 27/02/2018 11:54:08 AM Packet between 2 trays after Secondary Repeated 10 Times, From 120 Seconds. Sorter 27/02/2018 11:52:06 AM Packet between 2 trays after Secondary Repeated 10 Times, From 120 Seconds. Sorter 27/02/2018 11:52:05 AM Packet between 2 trays after Secondary Repeated 10 Times, From 120 Seconds. Sorter 27/02/2018 11:44:02 AM Packet between 2 trays after Secondary Repeated 10 Times, From 120 Seconds. Sorter 27/02/2018 11:43:01 AM Packet between 2 trays after Secondary Repeated 10 Times, From 120 Seconds.	Trolley[7932888704/0033] 2018 11:49:19 AM Extra Loop Taken, AWB+ 7932888704 and Extra Loop 0033 Repeated 5 Times. Feedline[2] 21/03/2018 11:41:52 AM Same NOBarcode Repeated 10 Times. Feedline[1] 21/03/2018 11:31:39 AM Same NOBarcode Repeated 10 Times. Feedline[2] 21/03/2018 11:24:34 AM Same NOBarcode Repeated 10 Times. Feedline[1] 21/03/2018 11:14:34 AM Same NOBarcode Repeated 10 Times. Station[16] 21/03/2018 9:35:55 AM Chute Detection Station[16] 21/03/2018 9:35:54 AM Chute Detection Station[7] 21/03/2018 9:35:53 AM Chute Detection

d. Training Portal

Customer Management

Product Management

Machine Management

Engineer & Training

Ticket & VM

Problem Management

Reports

Interest...

Dashboard

Master

Role Master

Department Master

Engineer Detail

Engineer Product Group

Training Type Master

Data Entry

Engineer Training

Training Material Master

#Home > Dashboard

Employee Training Admin

Add New

Employee Name Training Type Search Select Page Size

Show 10 Scheduled Completed

Employee Name Training Type Subject Status Training Start Date Actual Start Date Actual End Date Rating

Blank Page 01/11 Product Training TR Tray Sorter TR Tray Loop Sorter System Unit Chaudhary K.

Blank Page 01/11 Product Training System PTL Connected Belt Conveyor

Blank Page 01/11 Product Training BELT CONVEYOR WITH REACTION WHEEL Connected with TR Tray Sorter SHEEP

Blank Page 01/11 Product Training CONVEYOR SORTER SYSTEM TR Tray Sorter SHEEP

Blank Page 01/11 Product Training SORTER System System PTL_PU/PF SORTER FIL

Blank Page 01/11 Product Training VCL Cabinet Connected Check Hopper Belt Tact Check Hopper Cabinet Belt Conveyor

Blank Page 01/11 Product Training TR Tray Sorter SHEEP

Customer Management

Product Management

Machine Management

Engineer & Training

Ticket & VM

Problem Management

Reports

Interest...

Dashboard

Master

Role Master

Department Master

Engineer Detail

Engineer Product Group

Training Type Master

Data Entry

Engineer Training

Training Material Master

#Home > Dashboard

Training Material Master Admin

Add New

Training Type Document Name Video Name Search Select Page Size

Training Type Subject Document Name Training Document Video Name Training Video Time Limit Rating

Component Training KIT TRAINING SCFAC MANUAL download

Component Training WFO TRAINING POWERLIFT AIR MANUAL download 2

Component Training PTL PTL INSTRUCTION MANUAL download 2

Component Training PTL operation manual download 2

Product Training TR Tray Loop Sorter 1274 seconds setting manual download 2

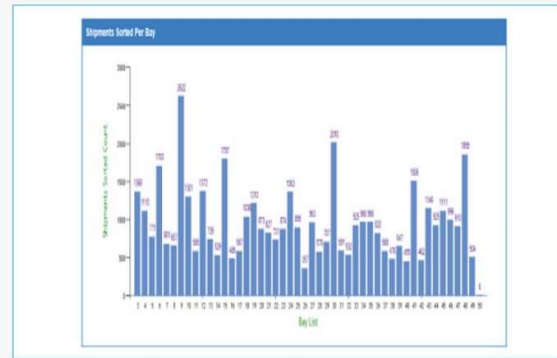
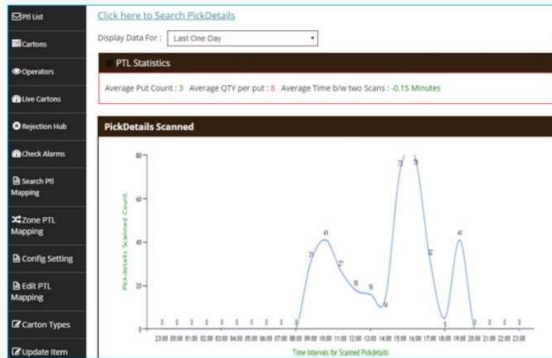
Maintenance & Troubleshooting TR Tray Sorter T214 MAINTENANCE MANUAL download 2

Product Training TR Tray Loop Sorter T214 OPERATION MANUAL download 2

Product Training TR Tray Loop Sorter T214 INDUCTION LINE MANUAL download 2

Department Procedures Field & Power Wiring Electrical-01 download 20 mins

e. Operational Analysis



21. SOPs for feeding

i. Initial Checks

1. Clean dust from sensors mounted on all conveyors and sorter (ensure that sensor should not be blinking or permanent on)
2. Remove all shipments from all conveyors
3. Ensure operators are at their designated stations only
4. Start machine by pressing START (green) button



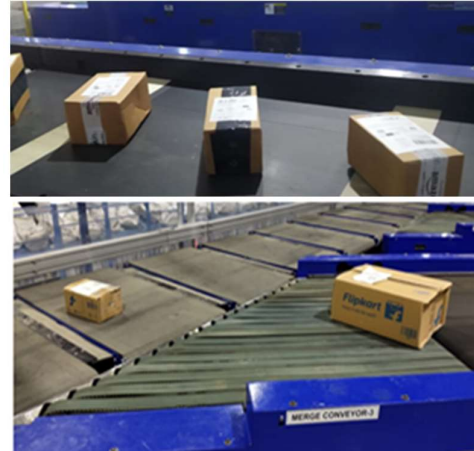
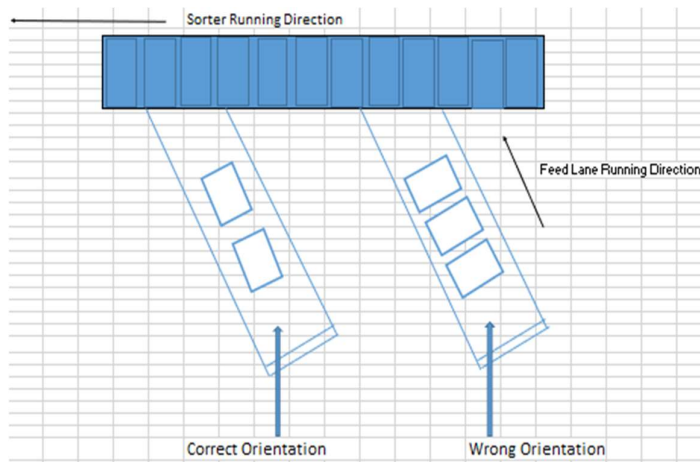
ii. Parcel Selection

1. Parcel size:
Minimum Parcel Size Should be 100(L) x 50 (W) x 5(H) mm
Maximum Parcel Size should be 600(L) x 400 (W) x 300(H) mm
2. Parcel weight: Max 10 KGs
3. Parcels with irregular base (unstable) should NOT be fed on to the sorter
4. Parcels with barcodes on the highest surface (max of L, B, H) should NOT be fed in automatic mode, as they tend to topple



iii.Feeding

1. Parcel should be placed on a flat base with barcode facing on the top side
2. While placing parcel on the loading conveyor, Longer edge of parcel should be perpendicular to the sorter movement direction (For alignment, use the 90-deg. wooden guide at the starting point of the conveyor, wherever provided)



3. If poly bag parcels have uneven distributed weight, then the heavier part should be oriented towards conveyor moving direction



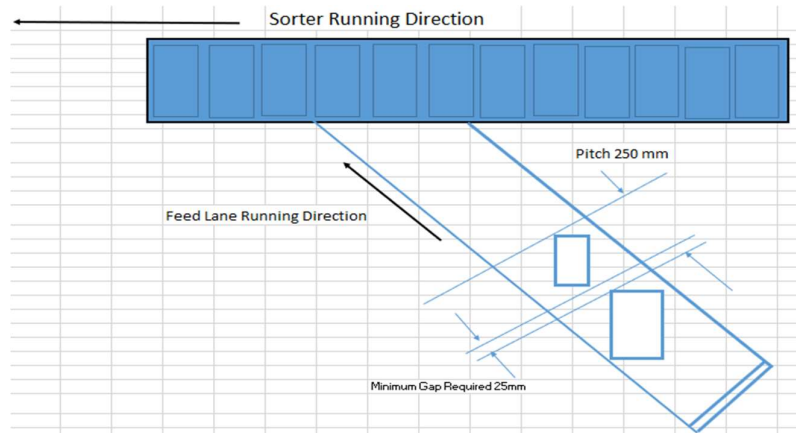
4. For poly packs / flyers, ensure the loose section of the poly is tightly wrapped around the solid section of the parcel to avoid double trigger to the sensor (Double trigger to sensor results in lesser throughput and false alarms)



5. Spacing between packets:

Smaller Parcels: - For products <250 mm length, the pitch of 250 mm between leading edge of adjacent packets to be maintained.

Larger Parcels: - For products >250 mm, a 25 mm gap between trailing edge and leading edge of adjacent parcels to be maintained



6. Overlapping parcels should NOT be fed on the sorter
7. Side-by-side parcels should NOT be fed on the sorter



22. Program Organisation

1. Program/ Project Schedule

Detailed Schedule shared in Technical Sheet

2. Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with Emirates SkyCargo
3. Scheduling of the resource management.
4. Management of the risks and opportunities.
5. Management of the requirements.
6. Management of the list of anomalies or reservations.

23. Price Schedule

Refer the file with Emirates SkyCargo 's Template

Total Price is to be understood

- Including Packing & Forwarding Charges
- Including Supervision Cost & Installation Charges
- Freight- Included
- Taxes Extra as applicable
- Unloading in Client's Scope
- Laydown Area to be provided by Emirates SkyCargo